

Jacob Wilson

DATA 5322 Practical Homework 3 Report

May 17, 2026

Convolution Neural Networks for Bird Species Classification

Abstract

The following report seeks to utilize machine learning models known as convolutional neural networks (CNNs) in order to identify bird species given sound clips of their calls. Two broad approaches were taken in this analysis: first classifying a clip as either belonging to a Bewick's Wren or Spotted Towhee (binary classification), and secondly by classifying between all 12 species available within the dataset (multi-class classification). In total five models were developed, 3 for binary classification and 2 for multi-class classification, each with their own differing parameters and underlying architectures. When evaluated on test data, the strongest binary classifier achieved an accuracy rate of 63.16% while the strongest multi-class classifier achieved an accuracy of 45.34%. Both multi-class models were then applied and evaluated with new, previously unseen audio clips in order to generate predictions of species.

Introduction

Machine learning provides the ability to interpret large datasets very quickly, with neural networks being among the best forms of machine learning models with widespread and increasing usage. Convolutional neural networks are a special family of neural networks that provide the ability to classify images, often mimicking human processing of images by first recognizing broad patterns before coalescing these patterns into a formal image classification (James, 2023, p. 407). Though audio classification is traditionally reserved for recurrent neural networks (RNNs), convolutional neural networks present an interesting alternative for bird call data, specifically when visualized in the form of a spectrogram.

The dataset utilized for this analysis was a preprocessed set containing audio clips of bird calls from 12 different species. The original data was gathered by the Cornell Lab of Ornithology's Center for Conservation Bioacoustics. The Center's data was compiled and made available by Xeno-Canto, an online archive for various animal sounds, via Kaggle (Rao, 2020). While initially containing roughly 36 gigabytes of data for a variety of bird species, this data was preprocessed to only contain data from 12

species, before selecting for 3 second clips of high quality sound data that were all converted into spectrogram images. These images would provide the base data on which the neural networks could be trained and later assessed.

Theoretical Background

Convolutional Neural Networks (CNNs)

Neural networks are a class of machine learning models that operate using three major types of underlying 'layers' that convert a given input into an output. These layers sequentially include the input layer, hidden layer, and output layer, with the input and output layers enacting some form of linear transformation on the data that they receive. By contrast, the defining feature of neural networks is the hidden layer, in which a nonlinear 'activation function' is applied to the data the layer receives. This nonlinear transformation, in particular, allows for a neural network to handle much more complex forms of data, including images, text, and audio. The name 'neural network' itself derives from viewing this architecture as resembling and mimicking neurons in the human brain (James, 2023, p. 401).

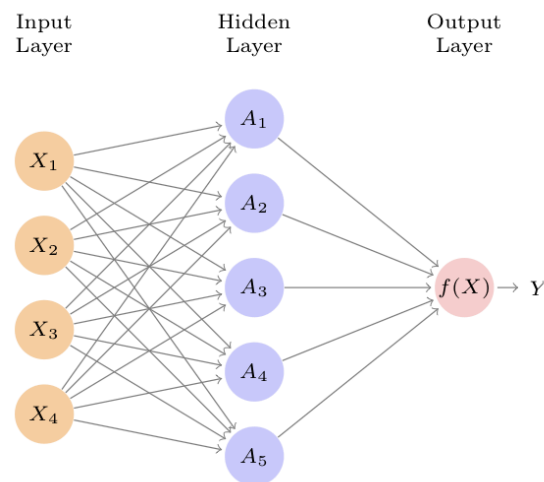


Figure 1. Visualization of single layer neural network structure

Convolutional neural networks constitute a subset of neural networks that utilize special hidden layers known as convolution and pooling layers. These layers operate in sequence, with convolution layers identifying small patterns in data, before pooling these patterns into smaller summaries that ultimately allow for classification (James, 2023, p. 407). These neural networks are most often utilized for image classification given their ability to identify key patterns within the image data (convolution) before summarizing these patterns into a numerical format that can then be classified

(pooling). Spectrograms are 2-dimensional visualizations of sound data that can thus be implemented into convolutional neural networks for the purposes of classification.

Binary vs. Multi-class Classification with Convolutional Neural Networks

In developing a convolutional neural network for the purposes of choosing between one of two classes, the model will utilize a sigmoid activation function in its output layer. This function will provide a linear transformation that will generate a series of probabilities between 0 and 1 in order to identify either class of interest. Additionally, these models rely on a binary cross-entropy function to evaluate loss, or the difference between the true class (0 or 1) and the probabilities generated by the model. Multi-class classification models, however, utilize a softmax activation function in their output layer in order to generate predictions between more than 2 classes. These models utilize categorical cross-entropy functions to evaluate loss. In either formulation, the loss is intended to be minimized in order to generate more accurate predictions of class labels (James, 2023, p.405).

Methodology

Data Preparation

The spectrogram images were developed by first subsampling each audio clip to a frequency of 22,050 Hz before shortening the clip to 3 seconds. From there, each 3 second clip was converted into a spectrogram comparing changes in sound frequency over time, with dimensions of 128 (frequency) by 517 (time). The spectrograms for each bird species were then stored as arrays with 3 dimensions: frequency, time, and the number of samples per species. In order to be interpreted by the convolutional neural network models, which require 4-dimensional input data, a fourth dimension of value 1 was appended to the array.

With the spectrograms converted into a new format, all of the data for each species of interest (2 for binary, 12 for multi-class) were amalgamated into a collective array that would act as the input. For the output, an array was generated where the data for each species was given numerical class labels. The input array was then normalized using z-score normalization in order to further scale down the input data and speed up the model processing. From this, the data was split with 80% of data points being reserved for model training while the other 20% of data would act as a test set to evaluate each model's performance.

It is worth noting that among the 1,981 total spectrograms for all species, there was significant, observable imbalance between the species. In particular, the House Sparrow (houspa) accounted for ~32% of all observations, constituting a plurality within the data. On the lowest end, another species, the Northern Flicker (norfli), accounted for only around 2% of total spectrograms.

Models

For binary classification, three separate models were generated and evaluated for their overall performance relative to one another. Each model utilized the same underlying neural network architecture, though with stronger dropout regularization applied to the the second and third models in an attempt to verify how this would impact performance. When fitting the model to the training data, the first model underwent twice as many rounds of training, or epochs, compared to the second and third models. Additionally, the latter two models had their batch sizes reduced by half compared to the first model, with only 16 samples passed through the neural network with each epoch. The key difference in the third model was the usage of the Adam optimization function when applying the model to the training data.

With multi-class classification, two models were developed. These models similarly possessed the same underlying architecture, with a key difference in batch size when being fit to the training data. In particular, the first model passed 16 samples per epoch through its architecture while the second model passed 8 samples per epoch.

Results

Figure 2 provides a comparative view of the accuracy and error rates of each binary classification model, both during training and when assessed against the test data. Despite the reduction of the number of epochs and batch size, the models broadly struggled to generalize to unseen data, with the first model performing best with an accuracy rate of ~63%. The strong accuracy rates from the models' training phase likely indicates that the models overfit to this data and thus could not effectively predict on the test data.

Noticeably, however, there was a minor improvement between the second and third binary models, with the latter regaining some accuracy when evaluated against the test data. This is likely due to the presence of the Adam optimization function in place of the RMSProp optimization function which is broadly the default for convolutional neural networks.

Model	Training Accuracy	Training Error Rate	Test Accuracy	Test Error Rate	Key Parameters
Model 1	91.52%	8.48%	63.16%	36.84%	dropout rate = 0.4, epochs = 30, batch size = 32
Model 2	90.18%	9.82%	50.88%	49.12%	dropout rate = 0.5, epochs = 15, batch size = 16
Model 3	91.07%	8.93%	54.39%	45.61%	dropout rate = 0.5, epochs = 15, batch size = 16

Figure 2. Table comparison of each binary classification model

The model results for the multi-class classification models are conveyed in Figure 3. When assessing for between 1 of 12 total classes as opposed to 1 of 2, the models generated worse overall test accuracies relative to their training accuracies. However, it is noteworthy that reducing the number of samples passed through with each training epoch resulted in an improvement in test accuracy between the first and second models.

When examining the predicted class labels relative to the true class labels, Figure 4 shows that the House Sparrow (houspa) was correctly identified more than any other species. Additionally, several predicted responses mistakenly classified other species as House Sparrows. Overall, however, it is observable that many of the spectrograms were mistakenly identified across all species.

Model	Training Accuracy	Training Error Rate	Test Accuracy	Test Error Rate	Key Parameters
Model 1	87.50%	12.50%	42.82%	57.18%	dropout rate = 0.5, epochs = 15, batch size = 16
Model 2	87.75%	12.25%	45.34%	54.66%	dropout rate = 0.5, epochs = 15, batch size = 8

Figure 3. Table comparison of each multi-class classification model

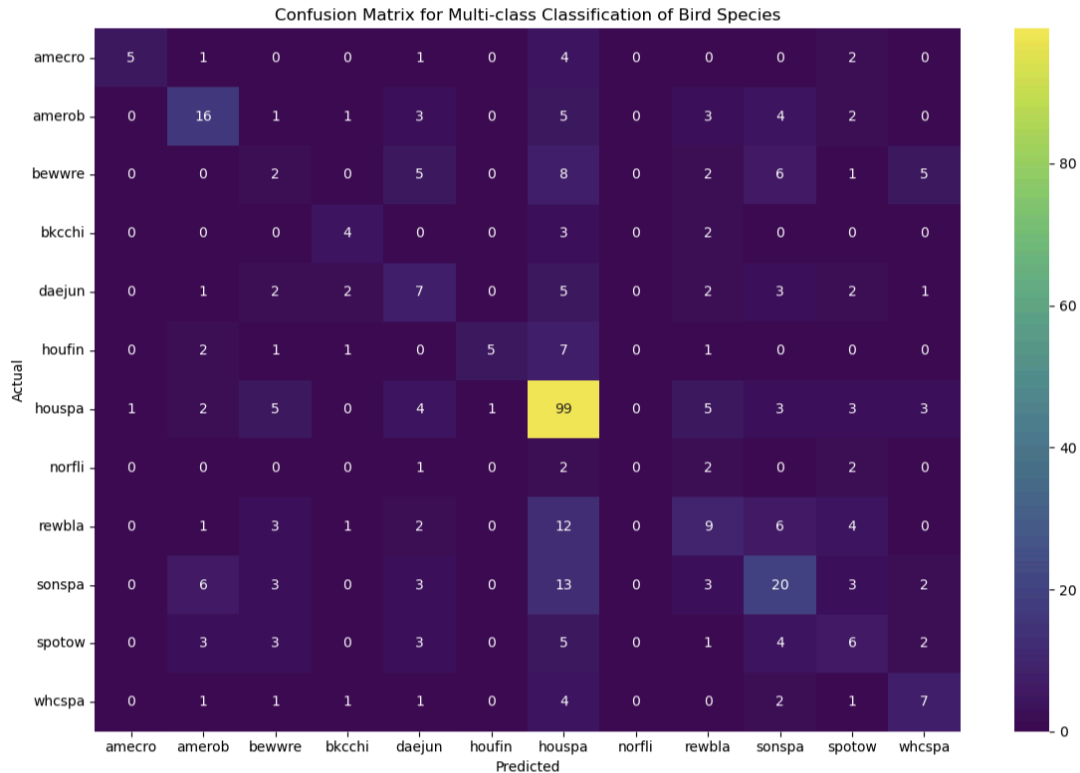


Figure 4. Confusion matrix comparing predicted species vs actual species for second multi-class classification model

When assessed on the three independent audio clips, the two multi-class models generated varying predictions with one another. When assessed on the first audio clip, both models identified the bird call as belonging to a House Sparrow, though with differing levels of probability: Model 1 generated a ~65% probability of the audio belonging to a House Sparrow while Model 2 had a 100% probability. For clips 2 and 3, however, Model 1 identified the calls as belonging to a Red-winged Blackbird while Model 2 again predicted House Sparrow.

Clip	Model 1 Prediction	Model 2 Prediction
Audio Clip 1	House Sparrow	House Sparrow
Audio Clip 2	Red-winged Blackbird	House Sparrow
Audio Clip 3	Red-winged Blackbird	House Sparrow

Figure 5. Table comparison of species classification for each of the three test audio clips

Discussion

Although the test accuracies of all five models were not particularly strong, the results do suggest that convolutional neural networks have the capability of identifying and learning patterns in spectrogram image data in order to generate classifications. The strongest overall model was the first binary classification model, which achieved a test accuracy of ~63%, suggesting some degree of learning from the training epochs that is beneficial over an arbitrary 50/50 prediction. This result may also imply that increased batch size and increased numbers of training epochs could provide a reasonable improvement in training accuracy with future models, though this will have to be further explored.

A significant limitation with this analysis was the consistent overfitting to the training data. This limitation was observed in all five models, and potentially suggests that the data might be too small relative to the number of underlying parameters of each model. As such, there is the potential that the models, with limited data to train on, may generalize very poorly to new data. Additional research could examine the usage of pre-trained models on the bird call data in order to compare results with the existing CNN models.

Another major limitation included the significant class imbalance between the 12 species in multi-class classification, which resulted in a strong bias towards predicting the majority class of House Sparrow relative to all other classes. This was further exemplified by testing the new audio clips on the multi-class classification models, with Model 2 predicting House Sparrow with near-absolute certainty. From the confusion matrices, several species' calls were also wrongly predicted as belonging to a House Sparrow, largely including those of Red-winged Blackbirds and Song Sparrows. This was also seen when applying the new audio clips to the multi-class models, where Models 1 and 2, respectively, identified clips 2 and 3 as belonging to a Red-winged Blackbird and a House Sparrow. When examining the waveforms and spectrograms of these clips, there was limited visual similarity between the three. However, this could suggest that there are less immediately observable patterns within the spectrograms that the neural networks are identifying. Additionally, there is also the potential for multiple birds to be uttering calls during the clips. In particular, the second test audio clip had a noticeably stronger amplitude compared to the others, which could be due to the presence of multiple bird calls at once.

Conclusions

The analysis conveys that convolutional neural networks indeed offer the potential of accurate classification when utilized on spectrogram image data. This was exemplified best by the first binary classification model, which achieved an overall test accuracy of 63.16%. While additional adjustments to the optimization function, dropout regularization, batch size, and number of training epochs broadly resulted in worse performance as the analysis proceeded, this does suggest that these parameters do have a strong impact on the ability of convolutional neural networks to generalize to new, unseen data.

Further attempts at applying these neural networks to spectrograms could account for class imbalance by potentially reducing the classes to be equal to one another, or by adjusting the aforementioned parameters. Additionally, the usage of pre-trained models on these spectrograms could provide a fascinating comparative analysis between complex, pre-trained neural networks and simpler neural networks.

Bibliography

James, G. Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An introduction to statistical learning: With applications in Python*. Springer.
<https://doi.org/10.1007/978-3-031-38747-0>

Rao, R. (2020). *Xeno-Canto bird recordings extended (A–M)* [Data set]. Kaggle.
<https://www.kaggle.com/datasets/rohanrao/xeno-canto-bird-recordings-extended-a-m>